Precision Medicine: A Machine Learning Approach to Diabetes Classification

Diabetes Classifier Application User Guide

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1. Introduction

Welcome to the Diabetes Classification dashboard. This tool will allow users to view different data visualizations, as well as three different machine learning classification models used to classify patients with or without diabetes.

Version: 3.1

Release Date: September 30, 2024

Support Contact: ewagner@my.gcu.edu

Getting Started

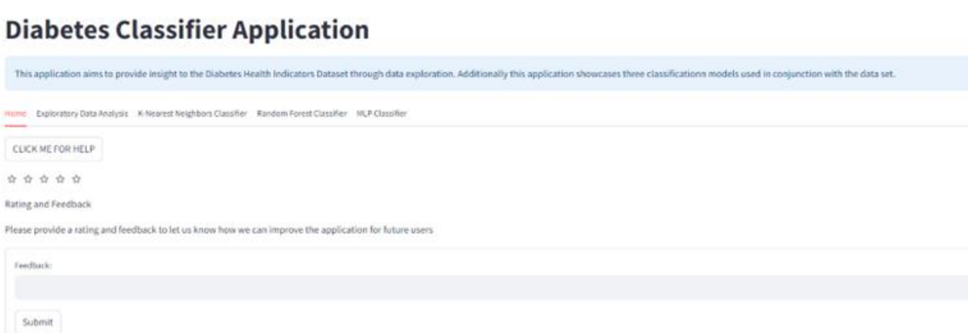
2.1 System Requirements

- Browser: Latest versions of Safari, Google Chrome, Microsoft Edge, or Fire fox
- Operating System: Current versions of Windows or macOS
- Internet Connection: Stable connection

2.2 Starting the API

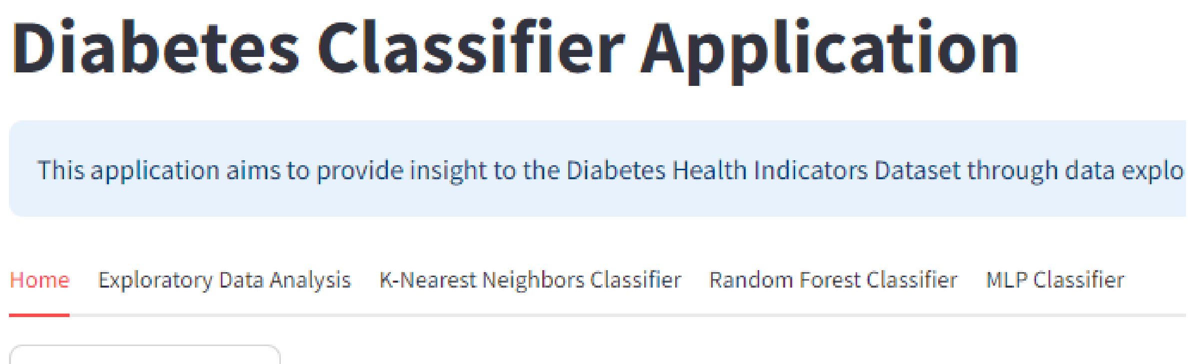
- Open the web browser and go to the Diabetes Classification API URL: https://diabestsclassappfinal-nvhdxq47v766q9s3wzurzq.streamlit.app/

- The API will load with the with home page:



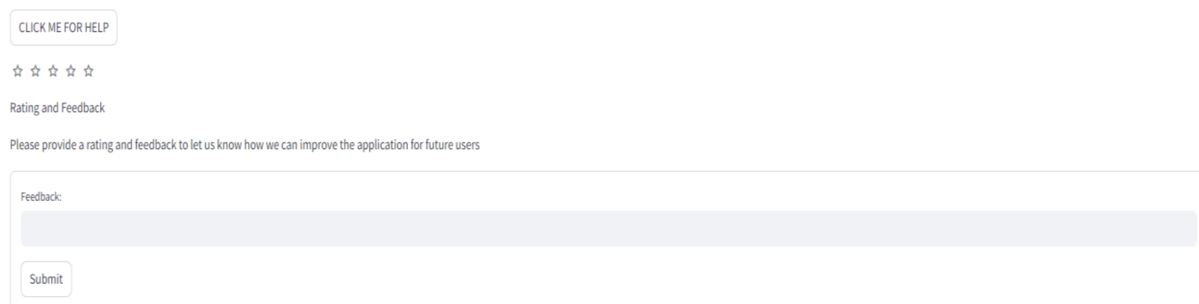
Dashboard Overview

2.2.1 *Site Tabs*

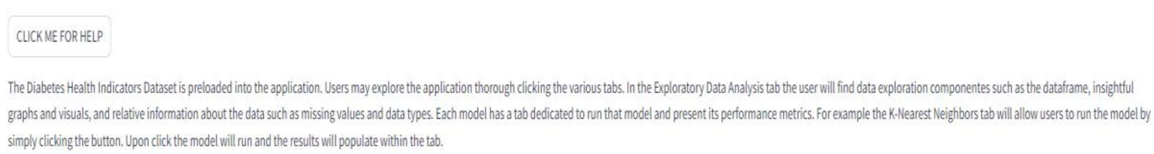


The API is designed with organization in mind, each section of the API is broken down into tabs. Fin explanations of each tab below:

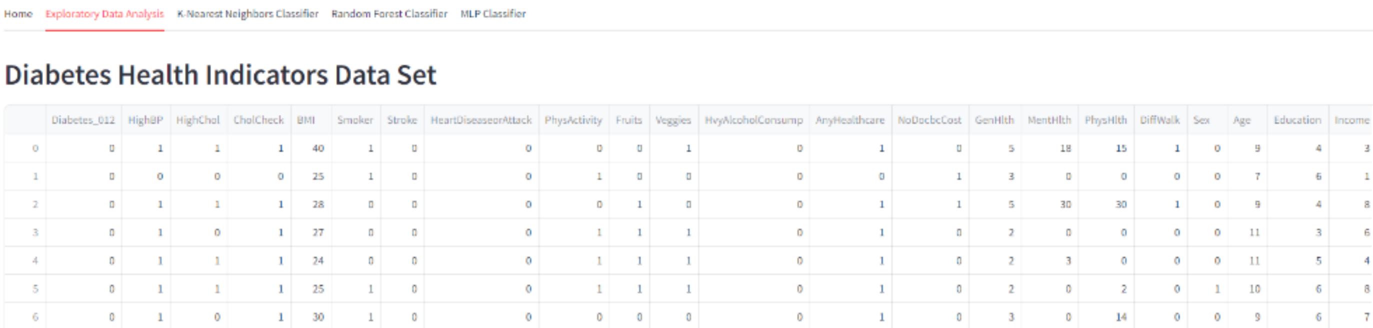
1) Home Tab



- Here users will find the initial help button, as well as a feature to rate the GUI and leave any feedback for the developer.
- Upon clicking the CLICK ME FOR HELP tab a brief paragraph summarizing the components of the API is displayed.



2) Exploratory Data Analysis Tab



Home | Exploratory Data Analysis | K-Nearest Neighbors Classifier | Random Forest Classifier | MLP Classifier

Diabetes Health Indicators Data Set

	Diabetes_012	HighBP	HighChol	CholCheck	BMI	Smoker	Stroke	HeartDiseaseorAttack	PhysActivity	Fruits	Veggies	HvyAlcoholConsump	AnyHealthcare	NoDocbcCost	GenHlth	MentHlth	PhysHlth	DiffWalk	Sex	Age	Education	Income
0	0	1	1	1	40	1	0	0	0	0	1	0	1	0	5	18	15	1	0	9	4	3
1	0	0	0	0	25	1	0	0	1	0	0	0	0	1	3	0	0	0	0	7	6	1
2	0	1	1	1	28	0	0	0	0	1	0	0	1	1	5	30	30	1	0	9	4	8
3	0	1	0	1	27	0	0	0	1	1	1	0	1	0	2	0	0	0	0	11	3	6
4	0	1	1	1	24	0	0	0	1	1	1	0	1	0	2	3	0	0	0	11	5	4
5	0	1	1	1	25	1	0	0	1	1	1	0	1	0	2	0	2	0	1	10	6	8
6	0	1	0	1	30	1	0	0	0	0	0	0	1	0	3	0	14	0	0	9	6	7

- This tab contains the Diabetes Health Indicators Dataset, a section with interactive buttons, interactive dropdowns, interactive dropdowns with visuals, and the final cleaned data set.

Diabetes Health Indicators Data Set Interactive Buttons

Data Shape

Null values within the Diabetes Health Indicators Dataset

Descriptive Statistics of original Diabetes Health Indicators Dataset

Number of duplicate values in the original Diabetes Health Indicators Dataset

Number of duplicate values in the cleaned Diabetes Health Indicators Dataset

Diabetes Health Indicators Data Set Interactive Drop-downs

Cleaned column names

Variable Data Types

Description of the variables found in the Diabetes Health Indicators dataset

Diabetes Class counts

Resampled Diabetes Class counts

Interactive Drop-Downs with visuals

Box plots of all features of the Diabetes Health Indicators Dataset

Boxplots with extreme outliers removed

Original Class Distribution

Class Distribution after Random Over-Sampling

Feature Selection via Correlation Matrix

Bar chart showing correlation between Diabetes Class and other variables(easier comprehension for non technical users)

Selected features and their correlation

Bar chart showing correlation between Diabetes Class and selected variables(easier comprehension for non technical users)

Bar charts used to show the density of important selected variables

Final Diabetes Health Indicators Dataframe

Cleaned, Balanced Data with feature selection applied

For security purposes all identifying patient information has been removed

	Diabetes_Class	High_BP	High_Chol	BMI	Smoker	Stroke	Heart_Disease_or_Attack	Phys_Activity_In_Last_30_Days	Hvy_Alcohol_Consump	Gen_Hlth	Ment_Hlth	Phys_Hlth	Diff_Walk	Sex	Age
1	0	0	0	25	1	0	0	1	0	3	0	0	0	0	7
3	0	1	0	27	0	0	0	1	0	2	0	0	0	0	11
4	0	1	1	24	0	0	0	1	0	2	3	0	0	0	11
5	0	1	1	25	1	0	0	1	0	2	0	2	0	1	10
7	0	1	1	25	1	0	0	1	0	3	0	0	1	0	11
9	0	0	0	24	0	0	0	0	0	2	0	0	0	1	8
10	2	0	0	25	1	0	0	1	0	3	0	0	0	1	13
13	2	1	1	28	0	0	0	0	0	4	0	0	1	0	11
15	0	1	0	33	0	0	0	1	0	2	5	0	0	0	6
16	0	1	1	21	0	0	0	1	0	3	0	0	0	0	10

i) The buttons on this page are:

- Data Shape

- Null values within the Diabetes Health Indicators Dataset
- Descriptive Statistics of the original Diabetes Health Indicators Dataset
- Number of Duplicated values in the original Diabetes Health Indicators Dataset
- Number of duplicate values in the cleaned Diabetes Health Indicators Dataset

ii) The Dropdowns on this page are:

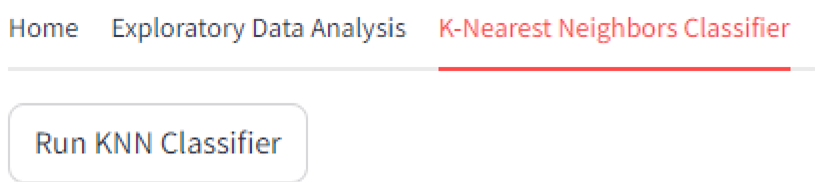
- Cleaned column names
- Variable Data Types
- Description of the variables found in the Diabetes Health Indicators dataset
- Diabetes Class counts
- Resampled Diabetes Class counts

Interactive Dropdowns with visuals

- Box plots of all features of the Diabetes Health Indicators Dataset
- Boxplots with extreme outliers removed
- Original Class Distribution
- Class Distribution after Random Over-Sampling
- Feature Selection via Correlation Matrix
- Bar chart showing correlation between Diabetes Class and other variables (easier comprehension for non-technical users)
- Selected features and their correlation

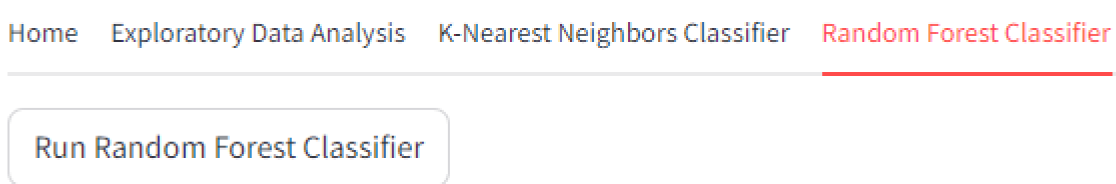
- Bar chart showing correlation between Diabetes Class and selected variables (easier comprehension for non-technical users)
- Bar charts used to show the density of important selected variables

2) K-Nearest Neighbors Classifier Tab



This tab is for the user to run the KNN Classifier and view its results
There is one button on the page: Run KNN Classifier

3) Random Forest Classifier



This tab is for the user to run the Random Forest Classifier and view its results
There is one button on the page: Run Random Forest Classifier

4) MLP Classifier Tab

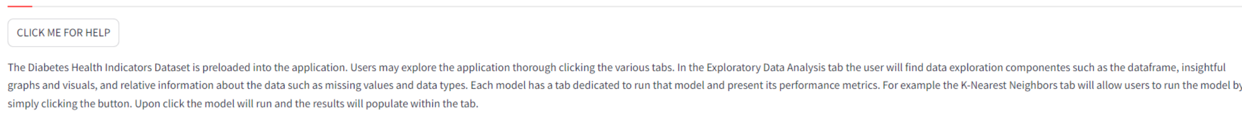
This tab is for the user to run the MLP Classifier and view its results
There is one button on the page: Run MLP Classifier

Using the Home Tab Dashboard

The Home tab is used to inform users about the application and provide feedback

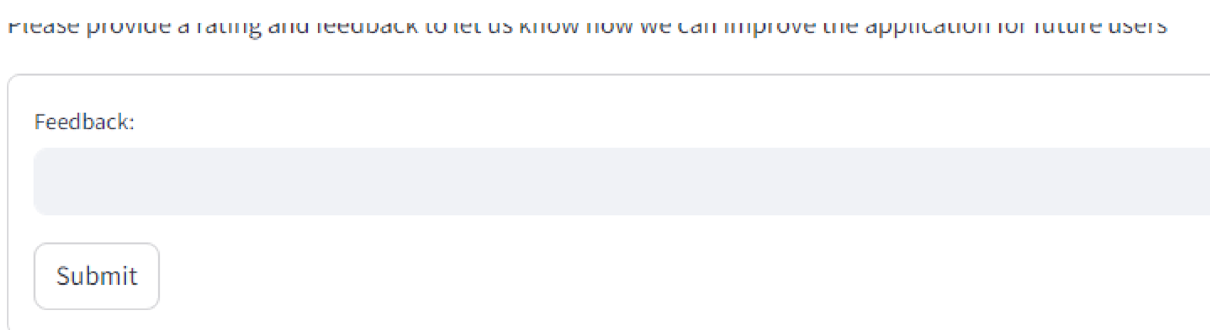
1) Click the “Click MR FOR HELP” button

(a) Upon clicking the button text will appear explain how to use the application



2. Using the stars the user may select a rating for the application
   - Clicking a star will cause it to light up yellow; the more stars are highlighted the better the rating

3. The feedback box is where users type any feedback for the application developer
   i. Click the gray box within the feedback box to type the feedback
   ii. Once ready click the submit button to send the feedback



Using the Exploratory Data Analysis Tab

i. The Exploratory Data Analysis Tab will load presenting the original unaltered data set.
ii. Below this dataset the application is broken into two columns with buttons on the left and dropdowns on the right

Diabetes Health Indicators Data Set Interactive Buttons:

- Upon clicking on the Data shape button, the shape of the data is displayed:

Data Shape

```
(253680, 22)
```

- Upon clicking on the “Null Values within the Diabetes Health Indicators Dataset: button, null values are displayed in a table:

Null Values within the Diabetes Health Indicators Dataset

	0
Diabetes_012	0
HighBP	0
HighChol	0
CholCheck	0
BMI	0
Smoker	0
Stroke	0
HeartDiseaseorAttack	0
PhysActivity	0
Fruits	0

- Upon clicking on the "Descriptive Statistics of the original Diabetes Health Indicators Dataset" button, the application will run a code to display the descriptive statistics in a table:

Descriptive Statistics of original Diabetes Health Indicators Dataset

	count	mean	std
Diabetes_012	253,680	0.2969	0.6982
HighBP	253,680	0.429	0.4949
HighChol	253,680	0.4241	0.4942
CholCheck	253,680	0.9627	0.1896
BMI	253,680	28.3824	6.6087
Smoker	253,680	0.4432	0.4968
Stroke	253,680	0.0406	0.1973
[illegible]	[illegible]	[illegible]	[illegible]

- Upon clicking on the "Number of duplicate Values in the original Diabetes Health Indicators Dataset" button, the application will display the number of duplicate entries:

Number of duplicate values in the original Diabetes Health Indicators Dataset

23899

- Upon clicking on the "Number of duplicate Values in the cleaned Diabetes Health Indicators Dataset" button, the application will display the number of duplicate entries:

Number of duplicate values in the cleaned Diabetes Health Indicators Dataset

0

Diabetes Health Indicators Data Set Interactive Dropdowns

Upon toggle of the “Cleaned column names” expander the cleaned column names are displayed:

Cleaned column names

Column names were cleaned for easier comprehension

	0
0	Diabetes_Class
1	High_BP
2	High_Chol
3	Had_Chol_Check_In_Last_5_Ye
4	BMI
5	Smoker

[redacted]

Upon toggle of the “Variable Data Types” expander the transformed variable type is displayed:

Variable Data Types

	0
Diabetes_Class	int64
High_BP	int64
High_Chol	int64
Had_Chol_Check_In_Last_5_Yea	int64
BMI	int64
Smoker	int64
Stroke	int64
Heart_Disease_or_Attack	int64
Phys_Activity_In_Last_30_Days	int64
Consumes_Fruits_Daily	int64

Upon toggle of the “Description of the variables found in the Diabetes Health Indicators dataset” expander a description of the features is displayed:

Description of the variables found in the Diabetes Health Indicators dataset

Description of features for easier comprehension of variables

	Description
Diabetes_Class	Diabetes status
HighBP	High blood pressure
HighChol	High cholesterol: (>24
CholCheck	Checked cholesterol i
BMI	Body mass index
Smoker	Smoked at least 100 c
Stroke	Had stroke or told by

Upon toggle of the “Diabetes Class counts” expander the original class counts are displayed:

Diabetes Class counts

Diabetes_Class	proportion
0	0.8271
2	0.1527
1	0.0201

Upon toggle of the “Resampled Diabetes Class counts” expander the resampled class counts are displayed:

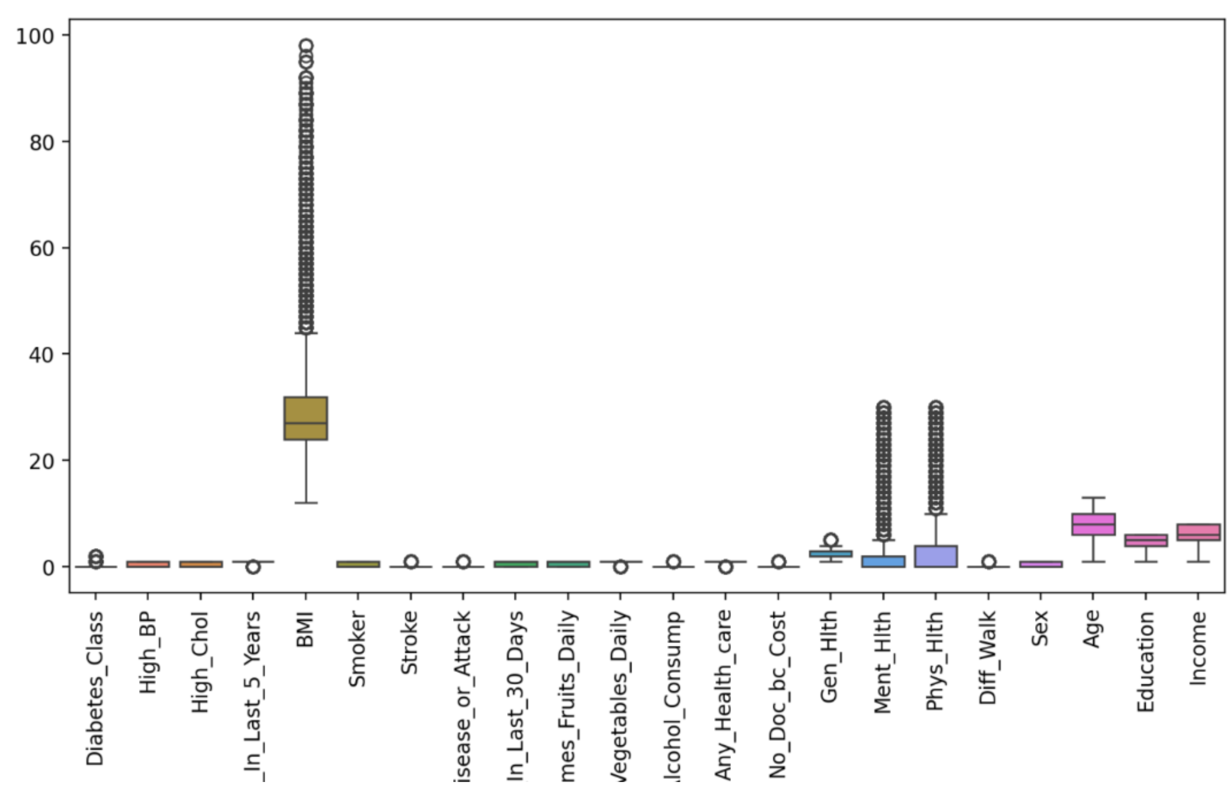
Resampled Diabetes Class counts

Diabetes_Class	proportion
0	0.3333
2	0.3333
1	0.3333

Interactive Dropdowns with visuals

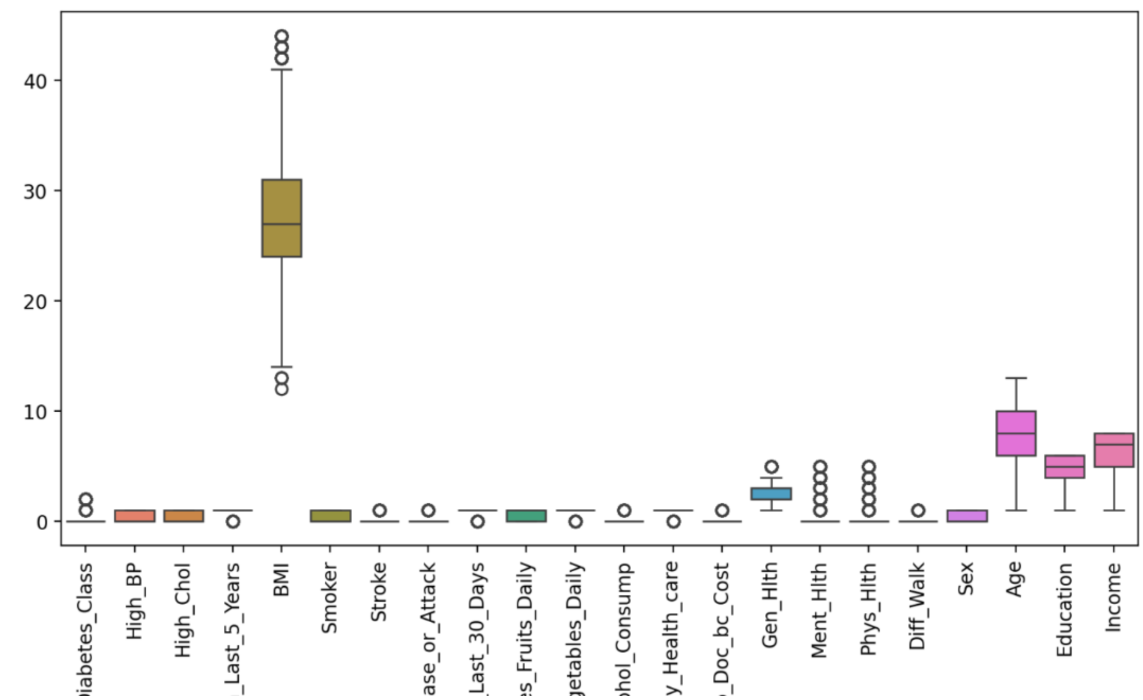
Upon toggle of the “Box plots of all features of the Diabetes Health Indicators Dataset” box plots are displayed for each variable:

Box plots of all features of the Diabetes Health Indicators Dataset



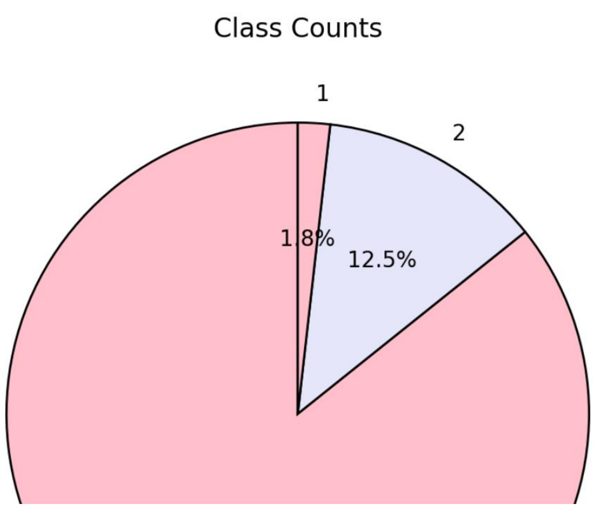
Upon toggle of the “Boxplots with extreme outliers removed” box plots are displayed for each variable with the outliers removed:

Boxplots with extreme outliers removed

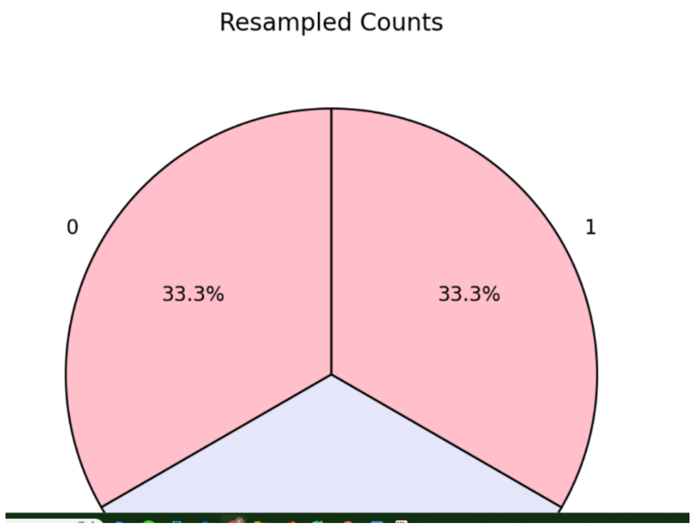


Upon toggle of the “Original Class Distribution” pie charts are displayed showing the original class distribution:

Distribution of classes before over sampling is used

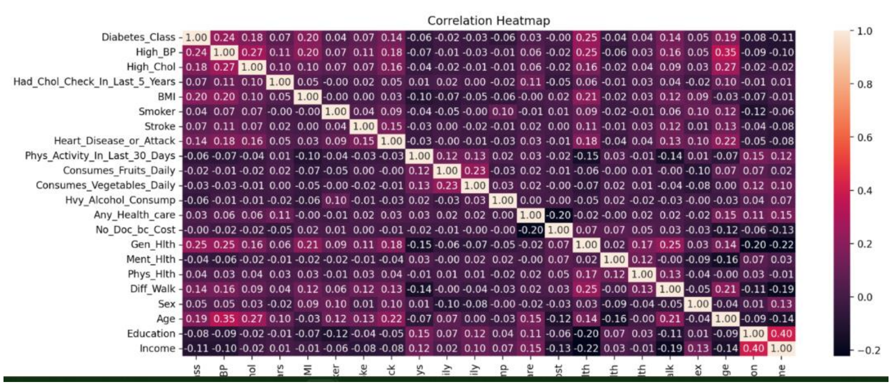


Upon toggle of the “Original Class Distribution” pie charts are displayed showing the original class distribution:

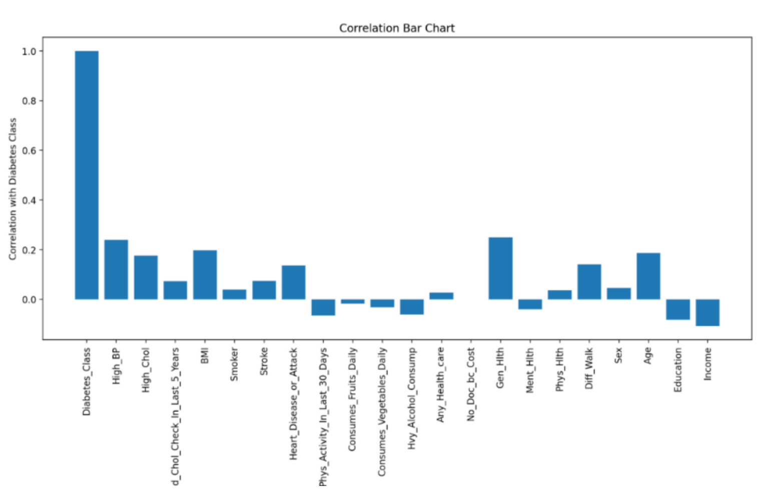


Upon toggle of the “Feature Selection via Correlation Matrix” expander a heat map is displayed showing the correlation of the variables:

Correlation matrix used to determine feature selection. Features are selected at with a value of .2 or greater, unless the variable was noted as an indicator by the WHO, CDC, or American Diabetes Association



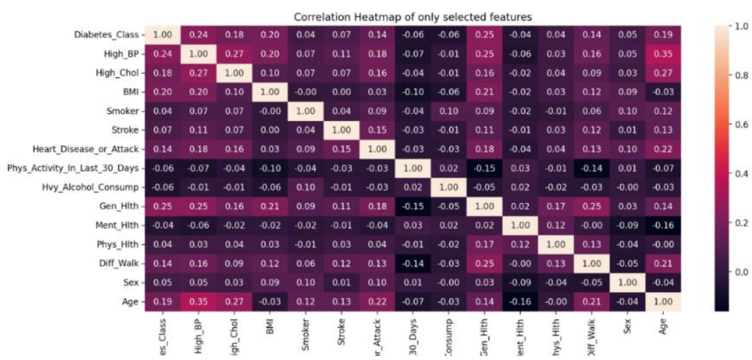
Upon toggle of the “Bar chart showing correlation between Diabetes Class and other variables (easier comprehension for non-technical users)” expander a bar chart is displayed showing the correlation of the variables:



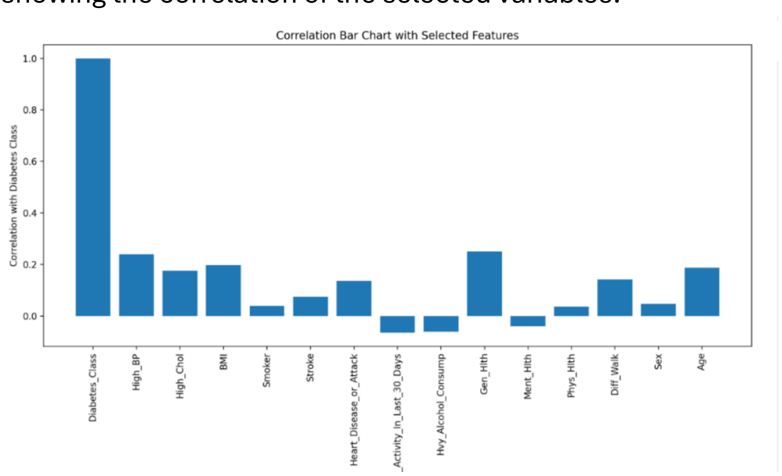
Upon toggle of the “Selected features and their correlation” expander a heat map is displayed showing the correlation of the variables:

Selected features and their correlation

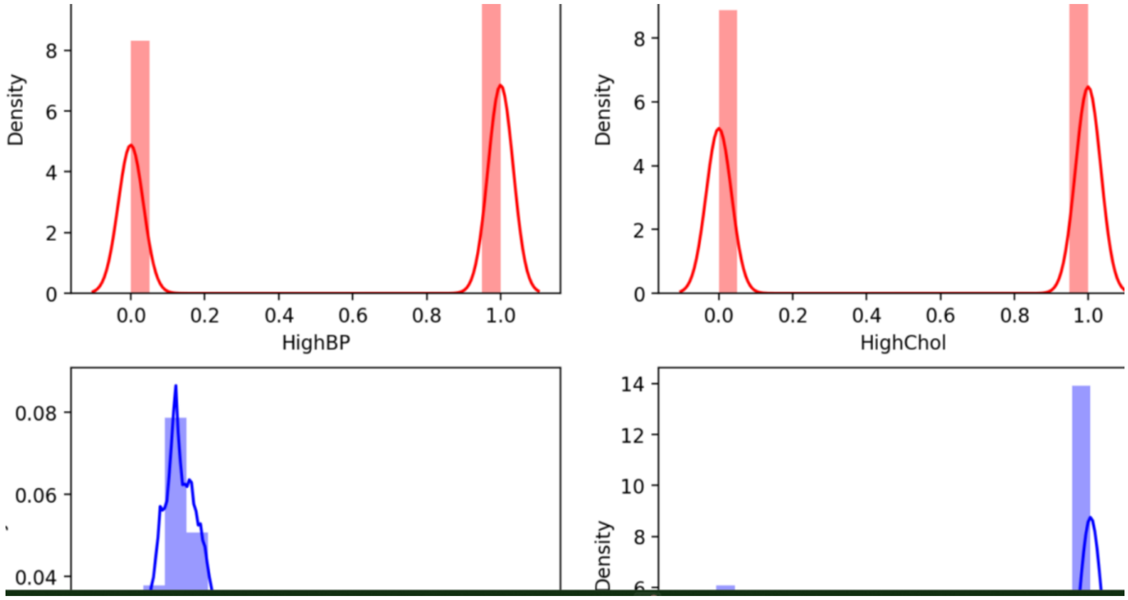
Correlation matrix to show the correlation of the selected features



Upon toggle of the “Bar chart showing correlation between Diabetes Class and selected variables (easier comprehension for non-technical users)” expander a bar chart is displayed showing the correlation of the selected variables:



Upon toggle of the “Bar charts used to show the density of important selected variables” expander bar charts are used to show the density of certain selected features:



Final Diabetes Health Indicators Dataframe

Upon toggle of the “Final Diabetes Health Indicators Data frame” expander the final clean data frame with feature selection applied is presented:

Cleaned, Balanced Data with feature selection applied

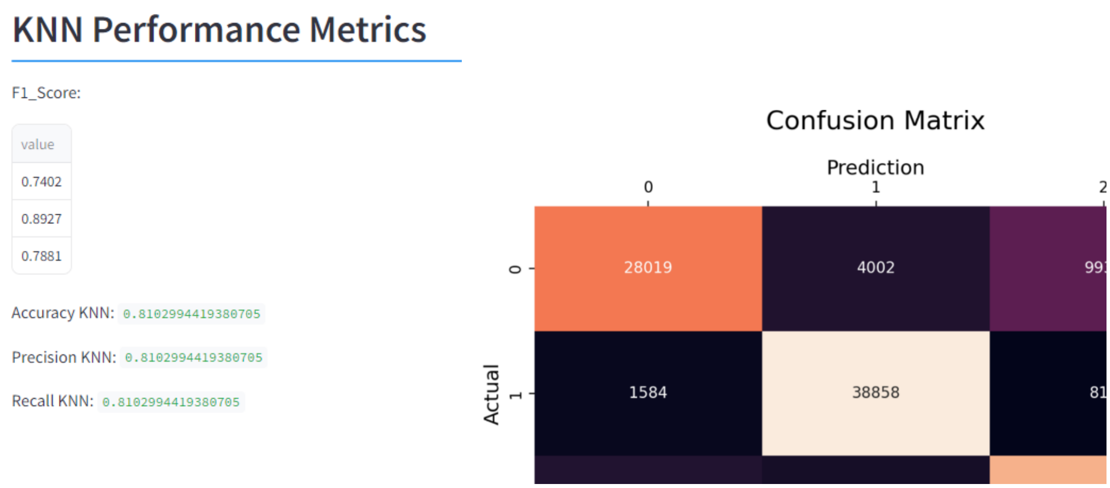
For security purposes all identifying patient information has been removed

	Diabetes_Class	High_BP	High_Chol	BMI	Smoker	Stroke	Heart_Disease_or_Attack	Pl
1	0	0	0	25	1	0	0	
3	0	1	0	27	0	0	0	
4	0	1	1	24	0	0	0	
5	0	1	1	25	1	0	0	
7	0	1	1	25	1	0	0	
9	0	0	0	24	0	0	0	
10	2	0	0	25	1	0	0	

Using the K-Nearest Neighbors Classifier Tab

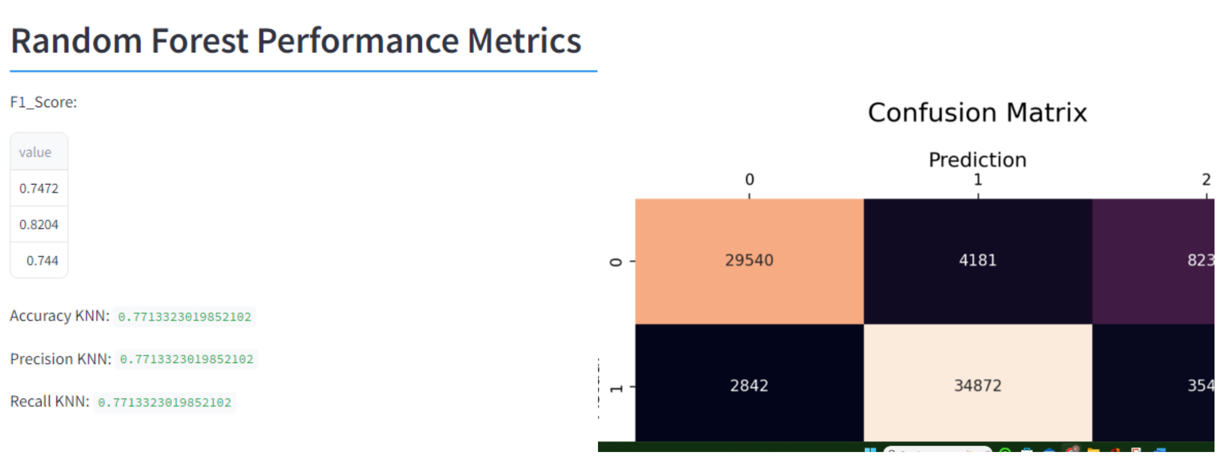
1. Upon clicking of the “Run KNN Classifier” button the application will run the KNN model and present the performance metrics:
   i. Accuracy
   ii. Precision

iii. Recall
iv. F-1 Score
v. Confusion Matrix



Using the Random Forest Classifier Tab

1. Upon clicking of the “Run Random Forest Classifier” button the application will run the Random Forest model and present the performance metrics:
   i. Accuracy
   ii. Precision
   iii. Recall
   iv. F-1 Score
   v. Confusion Matrix



Using the MLP Classifier Tab

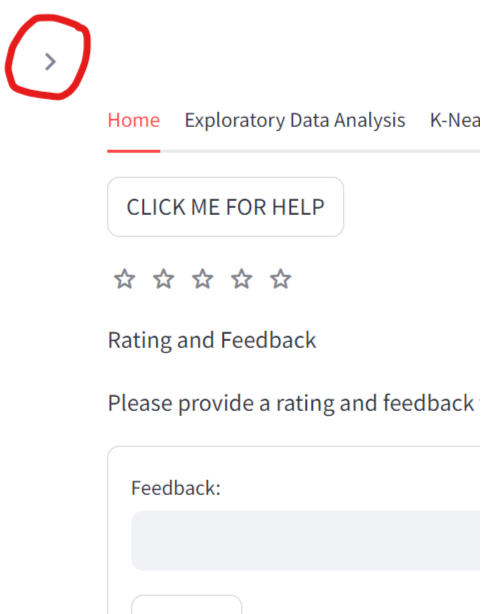
1. Upon clicking of the “Run MLP Classifier” button the application will run the MLP Classifier model and present the performance metrics:
   i. Accuracy
   ii. Precision
   iii. Recall
   iv. F-1 Score
   v. Confusion Matrix

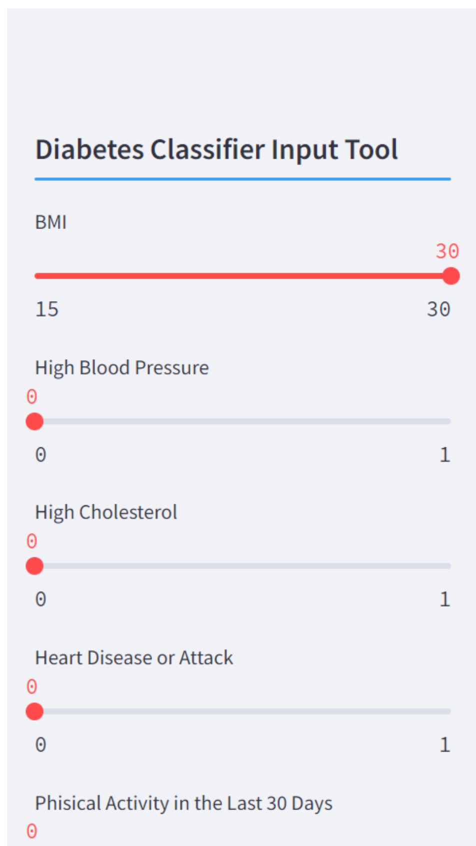
Validation

Using the Diabetes Classification Input tool

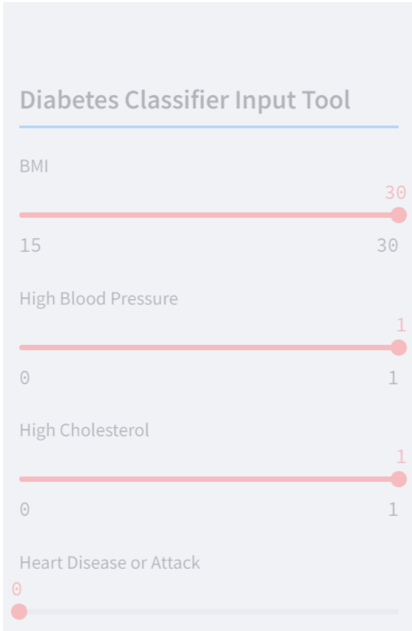
The sidebar allows users to input certain criteria then run the model to make a classification based on the input information.

1. Upon clicking the arrow located on the top left side of the screen a side bar opens



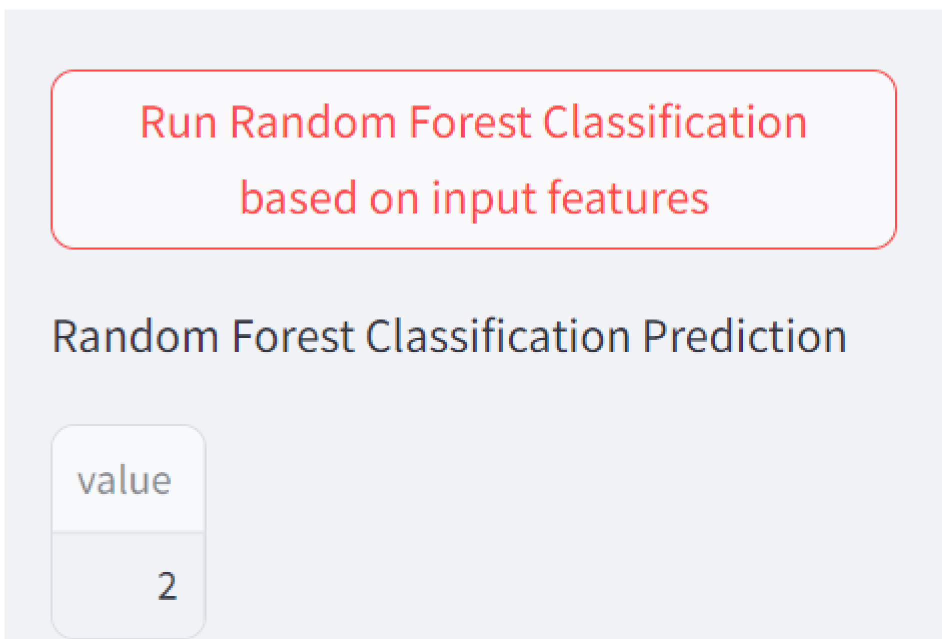


2. Upon clicking the circle located on the on the slider bar the user will drag the orange circle to the desired value

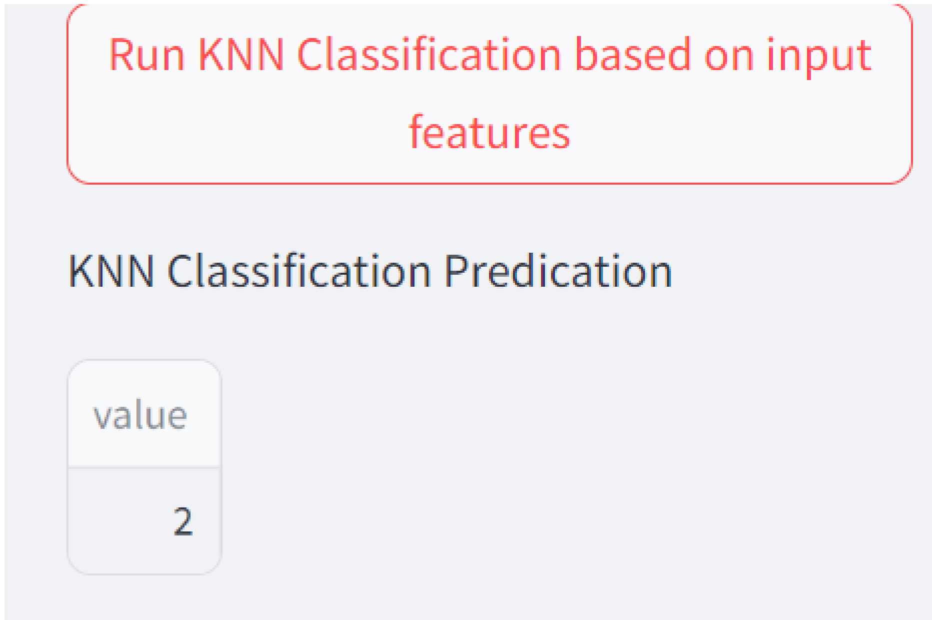


3. In the side bar there are thee buttons "Run Random Forest Classification based on input features", "Run KNN Classification based on input features", and "Run MLP Classification based on input features"

   iii. Upon clicking of the "Run Random Forest Classification based on input features" the model is run, and a classification is returned.



   iv. Upon clicking of the "Run KNN Classification based on input features" the model is run, and a classification is returned.



v. Upon clicking of the “Run MLP Classification based on input features” the model is run, and a classification is returned.

Frequently Asked Questions

1. Is the application able to show predictions given input data?
    a. Yes! By using the sidebar, the user can select certain input information and run the classification to get an output class
2. Can an updated data set be uploaded to the model?
    a. No, the current application is set to work and make classifications based on the 2015 BRFSS survey.

Help and Contact Details

- For help and other information regarding the Code or API please contact Emily Wagner at ewagner@my.gcu.edu